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Chapter 6

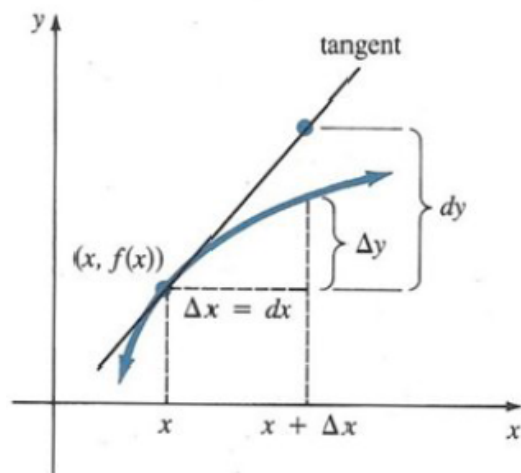
Applications of the Derivative

Gravitation cannot be held responsible for people falling in love. How on earth can you explain in terms of chemistry and physics so important a biological phenomenon as first love? Put your hand on a stove for a minute and it seems like an hour. Sit with that special girl for an hour and it seems like a minute. That's relativity.

—Albert Einstein

6.1 Approximation by Differentials

A method for approximating the value of a function near a known value. The method uses the tangent line at a known value of the function to approximate the function's graph. Let Δx and Δy represent the changes in x and y for the function and dx and dy represents the changes in x and y for the tangent line.



This is also written as

$$f(x + \Delta x) = f(x) + \Delta y = f(x) + f'(x)\Delta x.$$

Example: Approximate $\sqrt{10}$ by differentials.

Solution: $\sqrt{10}$ is near $\sqrt{9}$, so we will use $f(x) = \sqrt{x}$ with $x = 9$ and $\Delta x = 10 - 9 = 1$. Note that $f'(x) = \frac{1}{2\sqrt{x}}$. Therefore

$$\begin{aligned} \sqrt{10} &= f(x + \Delta x) \\ &\approx f(x) + f'(x)\Delta x \\ &= \sqrt{x} + \frac{1}{2\sqrt{x}}\Delta x = \sqrt{9} + \frac{1}{2\sqrt{9}}(1) \\ &= \frac{19}{6}. \end{aligned}$$

Example: Find an approximate value of $\sqrt[3]{9}$.

Solution: We set $f(x) = \sqrt[3]{x}$. Suppose to find $f(9)$. Note that $f'(x) = \frac{1}{3}x^{-\frac{2}{3}}$ and hence, $f(8) = 2$ and $f'(8) = \frac{1}{12}$, where $x = 8$ and $\Delta x = 9 - 8 = 1$. Therefore,

$$\begin{aligned} \sqrt[3]{9} &= f(9) \approx 2 + \frac{1}{12}(1) \\ &= \frac{25}{12} \\ &\approx 2.0833. \end{aligned}$$

Example: Find an approximate value for $\tan 46^\circ$.

Solution: Radial measure $46^\circ = 45^\circ + 1^\circ$ corresponds to $\frac{\pi}{4} + \frac{\pi}{180}$ and note that $f(x) = \tan x$, then

$f'(x) = \sec^2 x$ and $f(\frac{\pi}{4}) = 1$, $f'(\frac{\pi}{4}) = 2$. Therefore

$$\tan 46^\circ = \tan \left(\frac{\pi}{4} + \frac{\pi}{180} \right) \approx \tan \frac{\pi}{4} + \sec^2 \frac{\pi}{4} \left(\frac{\pi}{180} \right) = 1 + \frac{\pi}{90} \approx 1.0349.$$

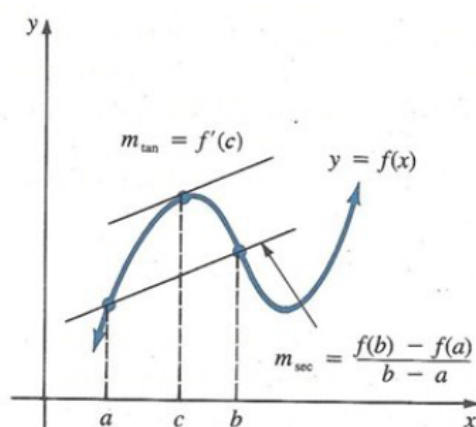
6.2 The Mean Value Theorem

Suppose $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) . Then, there exists a c in (a, b) at which the tangent line is parallel to the secant line joining the points $(a, f(a))$ and $(b, f(b))$, i.e., at which $f'(c) = \frac{f(b) - f(a)}{b - a}$,

OR

If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) , then there exists a point c in (a, b) such that

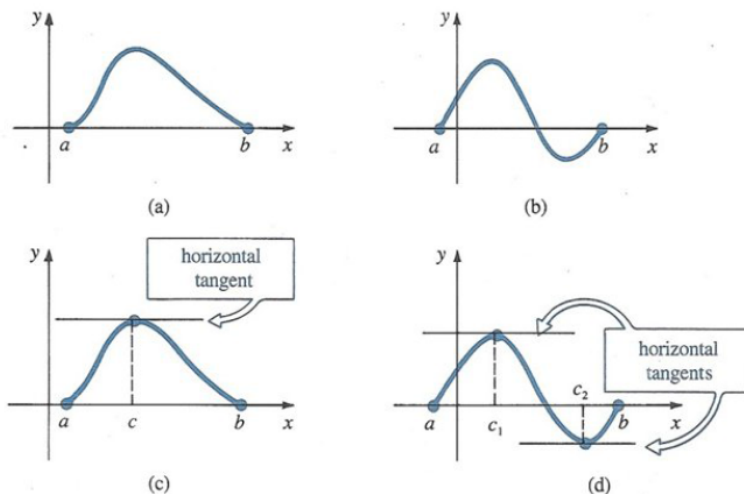
$$f'(c) = \frac{f(b) - f(a)}{b - a}, \quad a < c < b.$$



The word **mean** in The Mean Value Theorem refers to the mean (or average) rate of change of f in the interval $[a, b]$.

If $f(a) = f(b) = 0$, then the theorem says that there exists a c in (a, b) at which $f'(c) = 0$. The graphs suggest that there must be at least one point on the graph, that corresponds to a number c in (a, b) , at which the tangent is horizontal. This special case of the Mean Value Theorem is called **Rolle's Theorem**¹.

¹Michel Rolle, a French mathematician (1652-1719)



Example: Consider $f(x) = \sqrt{x-1}$ on $[2, 5]$, $f(x)$ is continuous when $x-1 \geq 0$, i.e., $x \geq 1$. In particular, $f(x)$ is continuous on $[2, 5]$ and $f'(x) = \frac{1}{2\sqrt{x-1}}$, so differentiable when $x > 1$. In particular, $f(x)$ is differentiable on $(2, 5)$.

$$\frac{f(b) - f(a)}{b - a} = \frac{f(5) - f(2)}{5 - 2} = \frac{\sqrt{5-1} - \sqrt{2-1}}{3} = \frac{1}{3}.$$

The Mean Value Theorem asserts that, for some c in $(2, 5)$, $f'(c) = \frac{1}{3}$. Let us find it.

$$\begin{aligned} f'(x) &= \frac{1}{3} \\ \frac{1}{2\sqrt{x-1}} &= \frac{1}{3} \\ 2\sqrt{x-1} &= 3 \\ 4(x-1) &= 9 \\ x-1 &= \frac{9}{4} \\ x &= \frac{13}{4}. \end{aligned}$$

Notice that $\frac{13}{4}$ is in $(2, 5)$, so we may take $c = \frac{13}{4}$.

Example: Show that if $f(x) = \tan x$ on the interval $0 \leq x \leq k$ where $k < \frac{\pi}{2}$, then $\tan k \geq k$.

Solution: By the Mean Value Theorem

$$\frac{\tan k - \tan 0}{k - 0} = \sec^2 c,$$

for some $c \in (0, k)$. But $\sec^2 c \geq 1$ and $\tan 0 = 0$. So

$$\frac{\tan k}{k} \geq 1 \implies \tan k \geq k.$$

Example: Use The Mean Value Theorem to show that $|\cos a - \cos b| \leq |a - b|$.

Solution: The function $\cos x$ is continuous and differentiable for all x . By the Mean Value Theorem

$$\begin{aligned}(\cos x)' &= \frac{\cos a - \cos b}{a - b} \\ |(\cos x)'| &= \left| \frac{\cos a - \cos b}{a - b} \right|,\end{aligned}$$

but $(\cos x)' = -\sin x$ and $|(\cos x)'| \leq 1$, therefore

$$\frac{|\cos a - \cos b|}{|a - b|} \leq 1 \implies |\cos a - \cos b| \leq |a - b|.$$

Example: Prove that $\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$ for $a < b$.

Solution: Let $f(x) = \tan^{-1} x$. Since $f'(x) = \frac{1}{1+x^2}$, $f'(c) = \frac{1}{1+c^2}$. By the Mean Value Theorem

$$\frac{f(b) - f(a)}{b - a} = \frac{\tan^{-1} b - \tan^{-1} a}{b - a} = \frac{1}{1+c^2}, \quad a < c < b.$$

Then, from $a < c < b$, we have

$$\begin{aligned}a^2 < c^2 < b^2 &\implies 1+a^2 < 1+c^2 < 1+b^2 \\ \frac{1}{1+a^2} > \frac{1}{1+c^2} > \frac{1}{1+b^2} &\implies \frac{1}{1+b^2} < \frac{1}{1+c^2} < \frac{1}{1+a^2} \\ \frac{1}{1+b^2} < \frac{\tan^{-1} b - \tan^{-1} a}{b - a} < \frac{1}{1+a^2} &\implies \frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}.\end{aligned}$$

Example: Use the Mean Value Theorem, to prove that if $0 < a < b$, then $1 - \frac{a}{b} < \ln \left(\frac{b}{a} \right) < \frac{b}{a} - 1$.

Hence show that $\frac{1}{6} < \ln 1.2 < \frac{1}{5}$.

Solution: Let $f(x) = \ln x$ and $f'(x) = \frac{1}{x}$. By the Mean Value Theorem, there exists $c \in (a, b)$ such that

$$f'(c) = \frac{1}{c} = \frac{\ln b - \ln a}{b - a}.$$

Then, from $a < c < b$ we have

$$\begin{aligned}a < c < b &\implies \frac{1}{b} < \frac{1}{c} < \frac{1}{a} \\ \frac{1}{b} < \frac{\ln b - \ln a}{b - a} < \frac{1}{a} &\implies \frac{b-a}{b} < \ln b - \ln a < \frac{b-a}{a} \\ &\implies 1 - \frac{a}{b} < \ln \left(\frac{b}{a} \right) < \frac{b}{a} - 1.\end{aligned}$$

Now, $\ln(1.2) = \ln\left(\frac{12}{10}\right) = \ln\left(\frac{6}{5}\right)$. Therefore $a = 5$ and $b = 6$. Substituting in $1 - \frac{a}{b} < \ln\left(\frac{b}{a}\right) < \frac{b}{a} - 1$, we have

$$1 - \frac{5}{6} < \ln\left(\frac{6}{5}\right) < \frac{6}{5} - 1 \implies \frac{1}{6} < \ln 1.2 < \frac{1}{5}.$$

6.3 Some Corollaries of The Mean Value Theorem

Corollary 6.3.1. *If $f'(x) = 0$ at all points of the interval (a, b) , then $f(x)$ must be a constant in the interval.*

Proof. Let $x_1 < x_2$ be any two different points in (a, b) . By the Mean Value Theorem for $x_1 < x < x_2$,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(x) = 0.$$

Thus $f(x_1) = f(x_2)$. Since x_1 and x_2 are arbitrarily chosen, the function $f(x)$ has the same value at all points in the interval. Thus, $f(x)$ is constant. $\square$

Corollary 6.3.2. *If $f'(x) > 0$ at all points of the interval (a, b) , then $f(x)$ is strictly increasing.*

Proof. Let $x_1 < x_2$ be any two different points in (a, b) . By the Mean Value Theorem for $x_1 < x < x_2$,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(x) > 0.$$

Thus $f(x_2) > f(x_1)$ for $x_2 > x_1$ and so $f(x)$ is strictly increasing. $\square$

6.4 Indeterminate Forms

Happens when $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ tends to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ as $x \rightarrow a$. Think of the situation $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \rightarrow \frac{0}{0}$ where $f(x)$ and $g(x)$ are differentiable (and therefore continuous so $f(a) = \lim_{x \rightarrow a} f(x) = 0$ and

$g(a) = \lim_{x \rightarrow a} g(x) = 0$), then

$$\begin{aligned}
 \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} \\
 &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad (\text{provided the denominator is not zero}) \\
 &= \lim_{x \rightarrow a} \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}} \\
 &= \frac{f'(a)}{g'(a)} = \frac{\lim_{x \rightarrow a} f'(x)}{\lim_{x \rightarrow a} g'(x)} \quad (\text{provided } f'(x) \text{ and } g'(x) \text{ are also continuous.})
 \end{aligned}$$

Example:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x^2 - 4)'}{(x - 2)'} = \lim_{x \rightarrow 2} \frac{2x}{1} = 2 \cdot 2 = 4.$$

Theorem 6.4.1. If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ provided that $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, this is called **L'Hôpital's Rule**: if either $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$.

Examples: (a) $\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 5x} = \lim_{x \rightarrow 0} \frac{2 \cos 2x}{5 \cos 5x} = \frac{2 \cdot 1}{5 \cdot 1} = \frac{2}{5}.$

(b) $\lim_{x \rightarrow \infty} \frac{e^{3x}}{x} = \lim_{x \rightarrow \infty} \frac{3e^{3x}}{1} = \infty.$

(c) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x^3} = \lim_{x \rightarrow 0} \frac{e^x}{3x^2} = \infty.$

The form $\infty - \infty$. A given limit that is not immediately $\frac{0}{0}$ or $\frac{\infty}{\infty}$ can be converted to one of these forms by combination of algebra and a little cleverness.

Example: Evaluate $\lim_{x \rightarrow 0} \left[\frac{1 + 3x}{\sin x} - \frac{1}{x} \right].$

Solution: We note $\frac{1 + 3x}{\sin x} \rightarrow \infty$ and $\frac{1}{x} \rightarrow \infty$. However, after writing the difference as a single

fraction, we recognize the form $\frac{0}{0}$.

$$\begin{aligned}\lim_{x \rightarrow 0} \left[\frac{1+3x}{\sin x} - \frac{1}{x} \right] &= \lim_{x \rightarrow 0} \frac{3x^2 + x - \sin x}{x \sin x} \\ &= \lim_{x \rightarrow 0} \frac{6x + 1 - \cos x}{x \cos x + \sin x} \\ &= \lim_{x \rightarrow 0} \frac{6 + \sin x}{-x \sin x + 2 \cos x} \\ &= \frac{6+0}{0+2} = 3.\end{aligned}$$

The form $0 \cdot \infty$. By suitable manipulation, L'Hôpital's Rule can sometimes be applied to the limit form $0 \cdot \infty$.

Example: Evaluate $\lim_{x \rightarrow \infty} x \sin \left(\frac{1}{x} \right)$.

Solution: Write the given expression as

$$\lim_{x \rightarrow \infty} \frac{\sin \left(\frac{1}{x} \right)}{\frac{1}{x}}$$

and recognize that we have the form $\frac{0}{0}$. Hence,

$$\begin{aligned}\lim_{x \rightarrow \infty} x \sin \left(\frac{1}{x} \right) &= \lim_{x \rightarrow \infty} \frac{(-x^{-2} \cos \frac{1}{x})}{(-x^{-2})} \\ &= \lim_{x \rightarrow \infty} \cos \frac{1}{x} = 1.\end{aligned}$$

The form 0^0 , ∞^0 , 1^∞ . Suppose $y = f(x)^{g(x)}$ tends towards 0^0 , ∞^0 , 1^∞ as $x \rightarrow a$ or $x \rightarrow \infty$. By taking the natural logarithm of y ;

$$\ln y = \ln f(x)^{g(x)} = g(x) \ln f(x)$$

and we see

$$\lim_{x \rightarrow a} \ln y = \lim_{x \rightarrow a} g(x) \ln f(x)$$

is of the form $0 \cdot \infty$. If it is assumed that $\lim_{x \rightarrow a} \ln y = \ln(\lim_{x \rightarrow a} y) = L$, then $\lim_{x \rightarrow a} y = e^L$ or

$$\lim_{x \rightarrow a} f(x)^{g(x)} = e^L.$$

Example: Evaluate $\lim_{x \rightarrow 0^+} x^{\frac{1}{\ln x}}$.

Solution: The form is 0^0 . Now, if we set $y = x^{\frac{1}{\ln x}}$, then

$$\ln y = \frac{1}{\ln x} \ln x = 1.$$

Notice we do not need L'Hôpital's Rule in this case since

$$\lim_{x \rightarrow 0^+} \ln y = 1.$$

Hence, $\lim_{x \rightarrow 0^+} y = e^1$ or equivalently $\lim_{x \rightarrow 0^+} x^{\frac{1}{\ln x}} = e$.

Example: Evaluate $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$.

Solution: The limit form is of the form 1^∞ . If $y = (1+x)^{\frac{1}{x}}$, then

$$\ln y = \frac{1}{x} \ln(1+x).$$

Now, $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$ has the form $\frac{0}{0}$ and so

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} &= \lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1} \\ &= \lim_{x \rightarrow 0} \frac{1}{1+x} = 1. \end{aligned}$$

Thus,

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e.$$

Example: Evaluate $\lim_{x \rightarrow \infty} \left(1 - \frac{3}{x}\right)^{2x}$.

Solution: The limit form is 1^∞ . If $y = \left(1 - \frac{3}{x}\right)^{2x}$ then $\ln y = 2x \ln \left(1 - \frac{3}{x}\right)$. Observe that the form $\lim_{x \rightarrow \infty} 2x \ln \left(1 - \frac{3}{x}\right)$ is $\infty \cdot 0$, whereas the form of $\lim_{x \rightarrow \infty} \frac{2 \ln(1 - \frac{3}{x})}{\frac{1}{x}}$ is $\frac{0}{0}$. Therefore,

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2 \ln(1 - \frac{3}{x})}{\frac{1}{x}} &= \lim_{x \rightarrow \infty} 2 \frac{\frac{\frac{3}{x^2}}{(1 - \frac{3}{x})}}{-\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{-6}{(1 - \frac{3}{x})} = -6. \end{aligned}$$

Finally, we conclude that

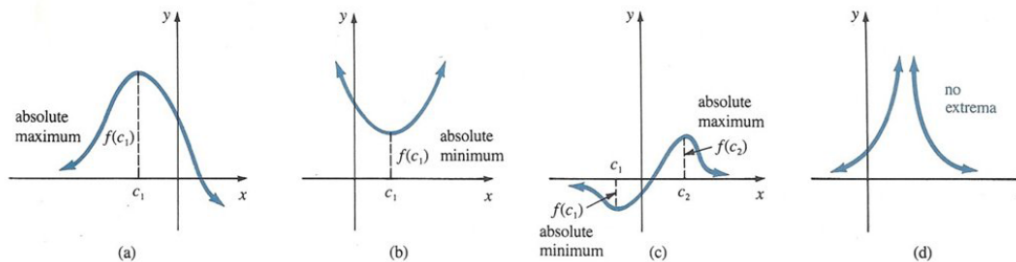
$$\lim_{x \rightarrow \infty} \left(1 - \frac{3}{x}\right)^{2x} = e^{-6}.$$

6.5 Extrema of Functions

Suppose a function f is defined on an interval I . The **maximum** and **minimum** values of f on I (if there are any) are said to be **extrema** of the function.

Definition 6.5.1. (i) A number $f(c_1)$ is an **absolute maximum** of a function f if $f(x) \leq f(c_1)$ for every x in the domain of f .

(ii) A number $f(c_1)$ is an **absolute minimum** of a function f if $f(x) \geq f(c_1)$ for every x in the domain of f .



Example: The function $f(x) = x^2$ has the absolute minimum $f(0) = 0$ but has no absolute maximum.

Example: $f(x) = \frac{1}{x}$ has neither an absolute maximum nor an absolute minimum.

The interval on which a function is defined is very important in the consideration of extrema.

Example: $f(x) = x^2$ defined only on the closed interval $[1, 2]$, has the absolute maximum $f(2) = 4$ and the absolute minimum $f(1) = 1$. On the other hand, if $f(x) = x^2$ is defined on the open interval $(1, 2)$, then f has no absolute extrema. In this case, $f(1)$ and $f(2)$ are not defined.

Theorem 6.5.1 (Extreme Value Theorem). A function f continuous on $[a, b]$ always has an absolute maximum and an absolute minimum on the interval.

Definition 6.5.2. (i) A number $f(c_1)$ is a **relative maximum** of a function f if $f(x) \leq f(c_1)$ for every x in some open interval that contains c_1 .

(ii) A number $f(c_1)$ is a **relative minimum** of a function f if $f(x) \geq f(c_1)$ for every x in some open interval that contains c_1 .

Definition 6.5.3. A **critical value** of a function f is a number c in its domain for which $f'(c) = 0$ or $f'(c)$ does not exist.

Example: Find the critical values of $f(x) = x^3 - 15x + 6$.

Solution:

$$f'(x) = 3x^2 - 15.$$

The critical values are those numbers for which $f'(x) = 0$, namely $\pm\sqrt{5}$.

Example: Find the critical value of $f(x) = (x + 4)^{\frac{2}{3}}$.

Solution: By power rule for functions,

$$f'(x) = \frac{2}{3}(x + 4)^{-\frac{1}{3}} = \frac{2}{3(x + 4)^{\frac{1}{3}}}.$$

In this instance we see that $f'(x)$ does not exist when $x = -4$. Since -4 is in the domain of f , we conclude it is a critical value.

Theorem 6.5.2. *If a function f has a relative extremum at a number c , then c is a critical value.*

6.6 Tutorial Questions

1. Evaluate each of the following limits using L'hôpital's rule.

$$\begin{aligned} & \text{(i)} \lim_{x \rightarrow 0} \frac{\sin x}{x} \quad \text{(ii)} \lim_{x \rightarrow 0} \frac{\sin x}{e^x - e^{-x}} \quad \text{(iii)} \lim_{x \rightarrow \infty} \frac{\ln x}{e^x} \quad \text{(iv)} \lim_{x \rightarrow \infty} \frac{6x^2 - 5x + 7}{4x^2 - 2x} \\ & \text{(v)} \lim_{x \rightarrow \infty} \frac{e^{3x}}{x^2} \quad \text{(vi)} \lim_{x \rightarrow \infty} \frac{x^4}{e^{2x}} \quad \text{(vii)} \lim_{t \rightarrow \frac{\pi}{4}^+} \frac{\tan t}{\tan 3t} \quad \text{(viii)} \lim_{x \rightarrow 1^+} \frac{\ln x}{\sqrt{x-1}} \\ & \text{(ix)} \lim_{x \rightarrow 0} \left[\frac{1+3x}{\sin x} - \frac{1}{x} \right] \quad \text{(x)} \lim_{x \rightarrow \infty} x \sin \left(\frac{1}{x} \right) \quad \text{(xi)} \lim_{x \rightarrow 0} x^{\frac{1}{\ln x}} \quad \text{(xii)} \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} \\ & \text{(xiii)} \lim_{x \rightarrow \infty} \left(1 - \frac{3}{x} \right)^{2x} \quad \text{(xiv)} \lim_{x \rightarrow 0^+} x^x \quad \text{(xv)} \lim_{x \rightarrow \infty} x(e^{\frac{1}{x}} - 1) \quad \text{(xvi)} \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right). \end{aligned}$$

2. Evaluate each of the following limits using L'Hôpital's rule.

$$\begin{aligned} & \text{(i)} \lim_{x \rightarrow \infty} (2 + e^x)^{e^{-x}} \quad \text{(ii)} \lim_{x \rightarrow 0^-} (1 - e^x)^{x^2} \quad \text{(iii)} \lim_{x \rightarrow \infty} x^2 e^{-x} \quad \text{(iv)} \lim_{x \rightarrow \infty} (x + e^x)^{\frac{2}{x}} \\ & \text{(v)} \lim_{x \rightarrow 0} x \ln(\sin x) \quad \text{(vi)} \lim_{x \rightarrow 0} x^{\ln^2 x} \quad \text{(vii)} \lim_{x \rightarrow 0} (1 + 5 \sin x)^{\cot x} \quad \text{(viii)} \lim_{x \rightarrow \infty} \left(\frac{3x}{3x+1} \right)^x \\ & \text{(ix)} \lim_{x \rightarrow 0} x^{(1-\cos x)} \quad \text{(x)} \lim_{t \rightarrow \infty} \left(1 + \frac{3}{t} \right)^t \quad \text{(xi)} \lim_{x \rightarrow 0} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right) \quad \text{(xii)} \lim_{x \rightarrow 0} \frac{\tan x}{2x} \\ & \text{(xiii)} \lim_{x \rightarrow 0^+} (1+x)^{\frac{1}{x^2}} \quad \text{(xiv)} \lim_{x \rightarrow \frac{\pi}{2}^+} (\sin^2 x)^{\tan x} \quad \text{(xv)} \lim_{\theta \rightarrow 0} \theta \csc 4\theta \quad \text{(xvi)} \lim_{x \rightarrow 0^+} x \ln x. \end{aligned}$$

3. Evaluate each of the following using L'hôpital's rule.

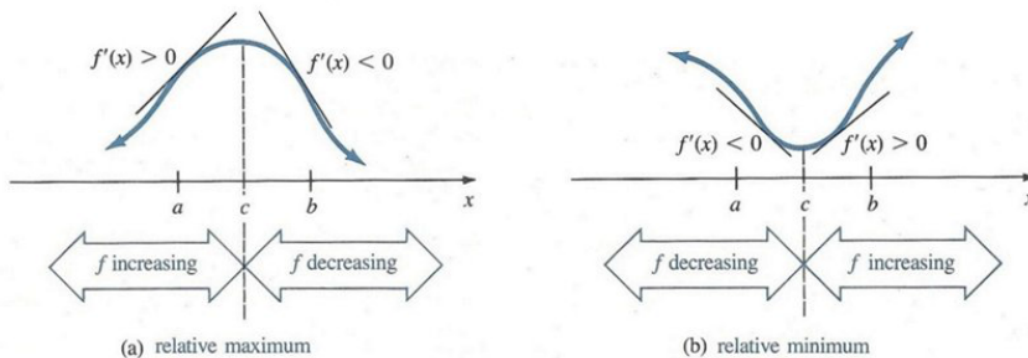
$$\begin{aligned} & \text{(i)} \lim_{r \rightarrow 0} \frac{r - \cos r}{r - \sin r} \quad \text{(ii)} \lim_{x \rightarrow \infty} \frac{1 + e^{-2x}}{1 - e^{-2x}} \quad \text{(iii)} \lim_{x \rightarrow 0} \frac{e^x - x - 1}{2x^2} \quad \text{(iv)} \lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x - \sin^{-1} x} \\ & \text{(v)} \lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} \quad \text{(vi)} \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} \right)^x \quad \text{(vii)} \lim_{x \rightarrow \infty} x^{\frac{1}{x}} \quad \text{(viii)} \lim_{x \rightarrow \frac{\pi}{4}} (\tan x)^{\tan x} \\ & \text{(ix)} \lim_{x \rightarrow 0} (x^2)^x \quad \text{(x)} \lim_{x \rightarrow 0} x^3 \ln x \quad \text{(xi)} \lim_{x \rightarrow 0} \frac{3^x - 2^x}{x} \quad \text{(xii)} \lim_{x \rightarrow 1^+} (\ln x)^{\ln x} \\ & \text{(xiii)} \lim_{x \rightarrow 0} x^{\sin x} \quad \text{(xiv)} \lim_{x \rightarrow \infty} (1+2x)^{\frac{1}{3x}} \quad \text{(xv)} \lim_{x \rightarrow 0} \left(\frac{e^x \sec x}{x} - \frac{\sin x}{x^2} \right) \quad \text{(xvi)} \lim_{x \rightarrow \infty} \frac{(x^2 + 3x + 3)^{x+1}}{(x+1)^{2x+1}(x+2)}. \end{aligned}$$

6.7 Graphing and the First Derivative

Knowing that a function does, or does not, possess relative extrema is a great aid in drawing its graph.

Theorem 6.7.1. Let f be continuous on $[a, b]$ and differentiable on (a, b) , except possibly at the critical value c .

- (i) If $f'(x) > 0$ for $a < x < c$ and $f'(x) < 0$ for $c < x < b$, then $f(c)$ is a **relative maximum**.
- (ii) If $f'(x) < 0$ for $a < x < c$ and $f'(x) > 0$ for $c < x < b$, then $f(c)$ is a **relative minimum**.
- (iii) If $f'(x)$ has the same algebraic sign on $a < x < c$ and $c < x < b$, then $f(c)$ is not an extremum.



Example: For each of the following functions, find all the critical points and classify each as a relative maximum, relative minimum or neither. (a) $f(x) = 3x^{\frac{5}{3}} - 15x^{\frac{2}{3}}$ and (b) $f(x) = x^3 - 3x^2 + 3x - 1$.

Solution: (a) Critical points:

$$\begin{aligned}
 f'(x) &= 5x^{\frac{2}{3}} - 10x^{-\frac{1}{3}} \\
 &= 5x^{-\frac{1}{3}}(x - 2) \\
 &= \frac{5(x - 2)}{x^{\frac{1}{3}}},
 \end{aligned}$$

which is zero when $x = 2$ and undefined when $x = 0$. Therefore $x = 0$ is a relative maximum and $x = 2$ is a relative minimum.

(b) Critical points:

$$\begin{aligned}
 f'(x) &= 3x^2 - 6x + 3 \\
 &= 3(x^2 - 2x + 1) \\
 &= 3(x - 1)^2,
 \end{aligned}$$

which is defined everywhere and zero when $x = 1$. Therefore $x = 1$ is neither.

Sometimes there is an easier way to test a critical point. Our goal is to relate the concept of the concavity of a graph with the second derivative of a function. Often a shape that is concave upwards is said to “hold water” whereas a shape that is concave downwards “spills water”.

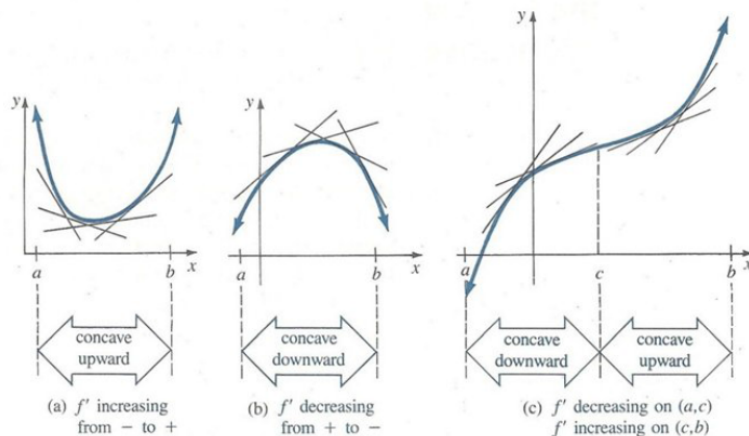
Definition 6.7.1. Let f be differentiable on (a, b) .

- (i) If f' is an increasing function on (a, b) , then the graph of f is **concave upwards** on the interval.
- (ii) If f' is a decreasing function on (a, b) , then the graph of f is **concave downwards** on the interval.



Theorem 6.7.2 (Test for Concavity). Let f be a function for which f'' exists on (a, b) .

- (i) If $f''(x) > 0$ for all x in (a, b) , then the graph is concave upward on (a, b) .
- (ii) If $f''(x) < 0$ for all x in (a, b) , then the graph is concave downward on (a, b) .



Example: Determine the intervals on which the graph of $f(x) = -x^3 + \frac{9}{2}x^2$ is concave upward and the intervals for which the graph is concave downward.

Solution: From

$$\begin{aligned}f'(x) &= -3x^2 + 9x \\f''(x) &= -6x + 9 = 6\left(-x + \frac{3}{2}\right),\end{aligned}$$

we see that $f''(x) > 0$ when $6\left(-x + \frac{3}{2}\right) > 0$ or $x < \frac{3}{2}$ and that $f''(x) < 0$ when $6\left(-x + \frac{3}{2}\right) < 0$ or $x > \frac{3}{2}$. It follows that the graph of f is concave upward on $(-\infty, \frac{3}{2})$ and concave downward on $(\frac{3}{2}, \infty)$.

6.7.1 Point of Inflection

Definition 6.7.2. Let f be continuous at c . A point $(c, f(c))$ is a **point of inflection** if there exists an open interval (a, b) that contains c such that the graph of f is either

- (i) concave upward on (a, c) and concave downward on (c, b) or
- (ii) concave downward on (a, c) and concave upward on (c, b) .

As a consequence, we observe that a point of inflection $(c, f(c))$ occurs at a number c for which $f''(c) = 0$ or $f''(c)$ does not exist.

Example: Find any points of inflection of $f(x) = -x^3 + x^2$.

Solution: The first and second derivatives of f are, respectively,

$$f'(x) = -3x^2 + 2x \quad \text{and} \quad f''(x) = -6x + 2.$$

Since $f''(x) = 0$ at $\frac{1}{3}$, the point $(\frac{1}{3}, \frac{2}{27})$ is the only possible point of inflection. Now,

$$\begin{aligned}f''(x) &= 6\left(-x + \frac{1}{3}\right) > 0 \quad \text{for } x < \frac{1}{3} \\f''(x) &= 6\left(-x + \frac{1}{3}\right) < 0 \quad \text{for } x > \frac{1}{3},\end{aligned}$$

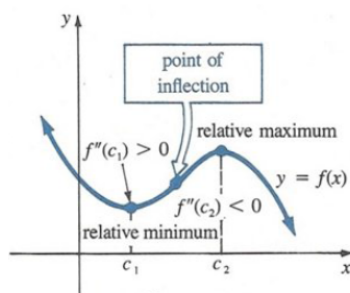
implies that the graph of f is concave upward on $(-\infty, \frac{1}{3})$ and concave downward on $(\frac{1}{3}, \infty)$. Thus $(\frac{1}{3}, f(\frac{1}{3}))$ or $(\frac{1}{3}, \frac{2}{27})$ is a point of inflection.

Second Derivative Test. If c is a critical value of $y = f(x)$ and, say, $f''(c) > 0$, then the graph of f is concave upward on some interval (a, b) that contains c . Necessarily then, $f(c)$ is a relative minimum. Similarly, $f''(c) < 0$ at a critical value c implies $f(c)$ is a relative maximum.

Theorem 6.7.3 (Second Derivative Test for Relative Extrema). Let f be a function for which f'' exists on an interval (a, b) that contains the critical number c .

(i) If $f''(c) > 0$, then $f(c)$ is a relative minimum.

(ii) If $f''(c) < 0$, then $f(c)$ is a relative maximum.



However, if $f''(c) = 0$ then nothing can be concluded about the nature of the critical point.

Example: for each of the following functions, find all of the critical points and classify each as relative maximum, relative minimum or neither. (a) $f(x) = x^3 - 3x + 2$ (b) $f(x) = \frac{1}{2}x - \sin x$ on $0 < x < 2\pi$.

Solution: (a)

$$\begin{aligned} f'(x) &= 3x^2 - 3 \\ &= 3(x^2 - 1) = 3(x - 1)(x + 1), \end{aligned}$$

which is defined everywhere and zero at $x = 1$ and $x = -1$. Computing $f''(x) = 6x$, then

$$f''(-1) = -6 < 0 \Rightarrow \text{relative maximum at } x = -1.$$

$$f''(1) = 6 > 0 \Rightarrow \text{relative minimum at } x = 1.$$

(b) $f'(x) = \frac{1}{2} - \cos x$ which is defined everywhere and zero when $\cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$. Computing $f''(x) = \sin x$, then

$$f''\left(\frac{\pi}{3}\right) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} > 0 \Rightarrow \text{relative minimum at } x = \frac{\pi}{3}.$$

$$f''\left(\frac{5\pi}{3}\right) = \sin \frac{5\pi}{3} = -\frac{\sqrt{3}}{2} < 0 \Rightarrow \text{relative maximum at } x = \frac{5\pi}{3}.$$

6.8 Sketching the graph of $y = f(x)$

6.9 Tutorial Questions

1. Use differentials to compute approximate value of

- (i) $\sqrt[3]{25}$ (ii) $\sin 31^\circ$ (iii) $\ln(1.12)$ (iv) $\sqrt[5]{36}$ (v) $\sqrt[5]{31}$ (vi) $\cos 62^\circ$

$$\begin{array}{llllll} \text{(vii)} \sin 48^\circ & \text{(viii)} (1005)^{\frac{1}{3}} & \text{(ix)} \tan 31^\circ & \text{(x)} \ln(1.2) & \text{(xi)} \sqrt{26} & \text{(xii)} \tan^{-1}(1.1) \\ \text{(xiii)} (1.1)^2 + 6(1.1)^2 & \text{(xiv)} \sin 1^\circ & \text{(xv)} \frac{1}{\sqrt{96}}. & & & \end{array}$$

2. Verify Rolle's Theorem for $f(x) = x^2(1-x)^2$, $0 \leq x \leq 1$.
3. Find the value of c in Rolle's Theorem when $f(x) = (x-a)^m(x-b)^n$ where m and n are positive integers.
4. If $f'(x) \leq 0$ at all points of (a, b) , prove that $f(x)$ is monotonic decreasing in (a, b) . Under what conditions is $f(x)$ strictly decreasing in (a, b) ?
5. Use the mean value theorem to show that $\sin x \leq x$ and $\tan x \geq x$. Hence show that $\frac{\sin x}{x}$ is strictly decreasing on $(0, \frac{\pi}{2})$.
6. Suppose f and g are continuous on $[a, b]$, differentiable on (a, b) and $f'(x) = g'(x)$ for all $x \in (a, b)$. Show that $f(x) = g(x) + c$, where c is a constant.
7. Use the mean value theorem, to prove that, if $0 < a < b$, then $(1 - \frac{a}{b}) < \ln(\frac{b}{a}) < (\frac{b}{a} - 1)$. Hence show that $\frac{1}{6} < \ln(1.2) < \frac{1}{5}$.
8. Show that if $0 < a < b$, then $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1}(b) - \sin^{-1}(a) < \frac{b-a}{\sqrt{1-b^2}}$. Hence show that $\left(\frac{\pi}{6} + \frac{\sqrt{3}}{15}\right) < \sin^{-1}(0.6) < \left(\frac{\pi}{6} + \frac{1}{8}\right)$.
9. Use the Mean Value Theorem to prove the following inequalities
 - (i) $\ln(1+x) < x$ if $x > 0$.
 - (ii) $\sqrt{1+x} < 1 + \frac{1}{2}x$ if $x > 0$.
 - (iii) $\frac{x-1}{x} < \ln x < x-1$ for $x > 1$.
10. Find the relative extrema of the given functions.
 - (i) $f(x) = 2x^3 + 3x^2 - 36x$
 - (ii) $f(x) = x^5 - \frac{5}{3}x^3 + 2$
 - (iii) $4x - 6x^{\frac{2}{3}} + 2$
 - (iv) $f(x) = \frac{x^2 - 2x + 2}{x-1}$.
11. Find the relative extrema and the points of inflection of the following functions.
 - (i) $f(x) = x^4 + 8x^3 + 18x^2$
 - (ii) $f(x) = x^6 - 3x^4 + 5$
 - (iii) $f(x) = 10 - (x-3)^{\frac{1}{3}}$
 - (iv) $f(x) = x(x-1)^{\frac{5}{2}}$.
12. Sketch the graphs of the following functions.
 - (i) $f(x) = 2x^2 - 5x - 3$
 - (ii) $f(x) = \frac{x+1}{(x-2)(3x+5)}$
 - (iii) $f(x) = x^4$
 - (iv) $f(x) = x^9$
 - (v) $f(x) = \frac{2x-1}{(x+2)(x-3)}$
 - (vi) $f(x) = \frac{x+1}{x-1}$
 - (vii) $f(x) = \left(\frac{x+1}{x-1}\right)^2$.